

Week 5 Capstone Progress Documentation

PW26_MG_02

Overview

During **Week 5**, we presented our **Phase 2 – Project Progress Review #1 (Review-1)** for our capstone project titled “**Toward Trustworthy Stock Prediction: A Self-Aware Multimodal Quant AI Framework.**” This review marked a key transition from conceptual groundwork to structured execution. The presentation covered the project motivation, scope, design philosophy, system architecture, and the planned roadmap for Phase 2.

Problem Motivation and Scope

We articulated the **problem statement** by highlighting the limitations of traditional quantitative and modern AI-based financial prediction systems, including rigidity, lack of transparency, limited accountability, and poor adaptability under volatile market conditions. Our project aims to address these challenges by building a **trustworthy, explainable, and self-aware AI-driven prediction framework** that emphasizes transparency and confidence-aware decision-making.

The **abstract and scope** highlighted the growing need for adaptive and auditable AI systems in regulated financial environments. We demonstrated how explainability, accountability, and adaptability together address investor distrust and regulatory expectations often unmet by opaque models.

Literature Insights and Research Gaps

Through the **literature survey**, we established that existing models perform adequately under stable conditions but struggle in dynamic regimes due to limited explainability and adaptability. This directly motivated our explainability-first architectural design. We identified key research gaps, including data provenance issues, lack of confidence estimation and contextual reasoning, absence of real-time adaptability, and weak accountability and auditability mechanisms.

Design Approach and Methodology

We presented a **frontend-first, modular, and polyglot design approach**, supported by UML diagrams and workflow visualizations. The core philosophy is to execute decisions only under high confidence with transparent justification and abstain under uncertainty to preserve system integrity, treating explainability as the execution gate.

Architecture and Phase 2 Planning

A layered system architecture was introduced, along with implementation challenges and resource requirements, particularly GPU acceleration. A structured Phase 2 progress plan was also outlined.

Panel Feedback and Reflection

The panelists appreciated the architectural clarity and explainability-driven design, recommending the development of an **explainability toolkit** in upcoming phases. They noted the project’s potential to **revolutionize the future of finance.**

Overall, Week 5 was an enriching experience that strengthened our direction for Phase 2.